

CO1 ASSESSMENT TOOL 2: CASE STUDY ANALYSIS

Smart High-Speed Rail Passenger Experience Platform

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1. Project Vision and Stakeholders

1.1 Project Vision

To create a unified, intelligent digital platform that transforms the high-speed rail passenger journey — from booking to arrival — by delivering real-time train information, seamless mobile ticketing, personalized in-journey services, and predictive assistance, thereby improving customer satisfaction, operational efficiency, and ridership on the high-speed rail network.

1.2 Project Objectives

- Provide real-time train updates, seamless QR ticketing, personalized onboard services, and improved accessibility for all passengers.
- Reduce station congestion, deliver proactive disruption alerts with alternatives, and increase non-fare revenue through digital services.

1.3 Key Stakeholders

Stakeholder	Role / Interest in the Project
Passengers (commuters, business travelers, tourists)	Primary end-users; need convenient booking, real-time information, and a comfortable journey.
Differently-abled / elderly passengers	Require accessible booking and assured onboard/station assistance.

Stakeholder	Role / Interest in the Project
Product Owner	Owns the product vision, prioritizes the backlog, and represents stakeholder interests to the team.
Scrum Master	Facilitates Agile ceremonies and removes impediments for the development team.
Onboard Staff (Train Captains, Catering Crew)	Fulfil in-journey requests such as food orders and assistance tasks.
Customer Support / Help Desk	Handle passenger queries and escalations raised through the in-app chat.

2. Epics and User Stories

The product requirements have been grouped into seven Epics. Representative user stories are written in the standard Agile format below.

Epic 1 — Smart Ticketing & Booking

Epic 2 — Real-Time Journey Information

Epic 3 — Personalized Passenger Experience

Epic 4 — Station & Crowd Management

Epic 5 — Onboard Services

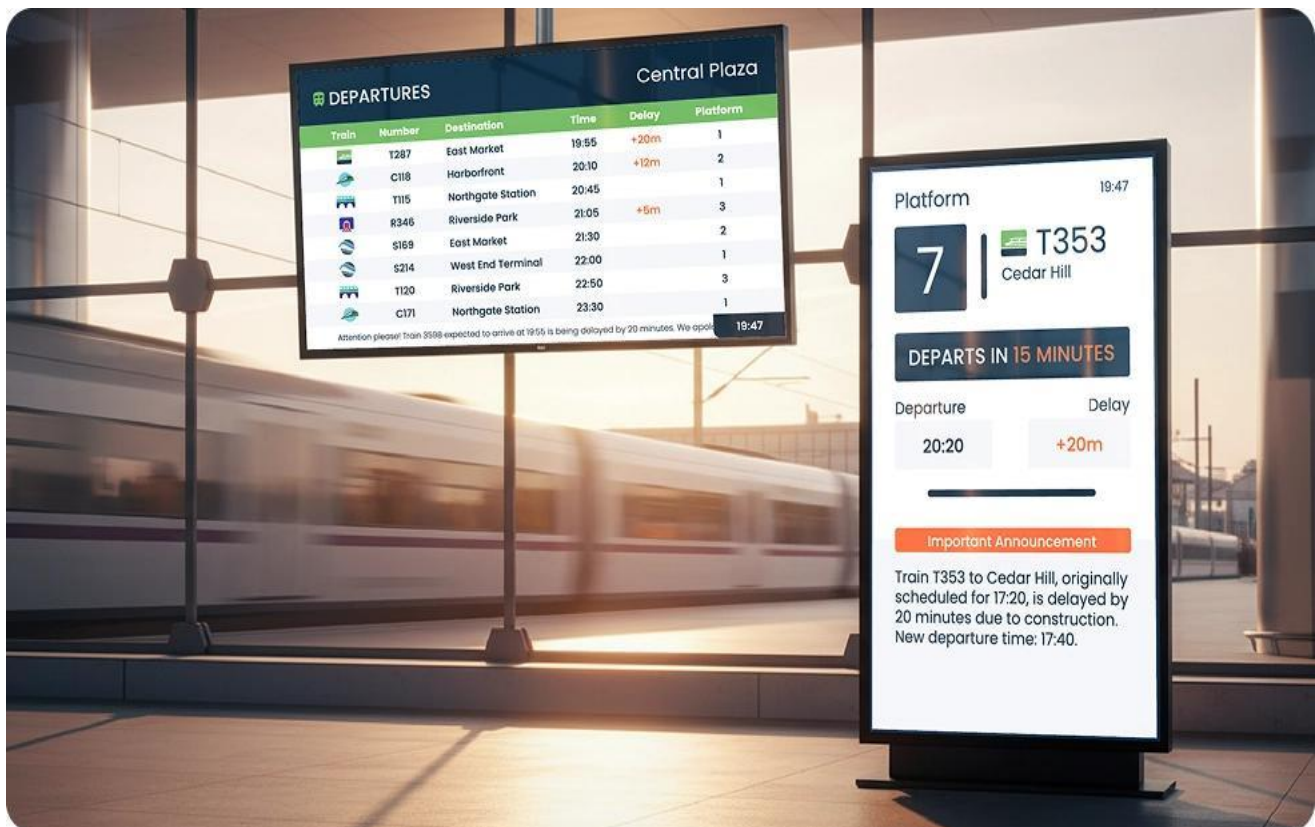
Epic 6 — Accessibility & Support

Epic 7 — Notifications & Disruption Management

3. Acceptance Criteria

Acceptance criteria are defined in Given–When–Then format for the highest-priority user stories selected for early delivery.

- Booking displays seats and fare, payment generates confirmation.
- QR e-ticket generated and delivered through SMS, email and app.
- Live tracking refreshes every 60 seconds.
- Platform notifications update automatically.
- Crowd alerts recommend staff deployment.



4. Product Backlog Prioritization

Rank	User Story	Priority	Justification
1	US-01 Book tickets via app	High	Core revenue-generating function; a dependency for almost every other feature.
2	US-03 QR e-ticket	High	Required for boarding; directly dependent on booking.
3	US-16 Delay notifications	High	High impact on customer satisfaction and trust; expected baseline service.
4	US-04 Live train tracking	High	Key experience differentiator; enables platform notification feature.
5	US-05 Platform notification	Medium-High	Builds on live tracking; strong day-to-day utility.

6. Story Point Estimation

User Story	Story Points	Justification
US-01 Book tickets via app	8	High complexity: search, seat inventory, UI, and a third-party payment integration.
US-03 QR e-ticket	5	Moderate complexity; depends on booking service, needs generation plus multi-channel delivery.
US-14 Wheelchair assistance	3	Simple workflow: request form plus staff task routing.
US-15 In-app support chat	5	Requires integration with a support/ticketing backend and real-time messaging.

User Story	Story Points	Justification
US-10 Footfall analytics	8	High complexity: sensor integration, dashboard, and threshold-based alerting.
US-12 Food ordering	5	Moderate: order flow, catalog, and routing to catering staff.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

“By the end of Sprint 1, passengers can search and book high-speed rail tickets through the mobile app, receive a QR-coded e-ticket, track their train in real time, and get instant delay notifications — establishing the core MVP passenger journey.”

7.2 Expected Sprint Deliverables

- A functional ticket booking module integrated with a payment gateway.
- QR-code e-ticket generation with SMS, email, and in-app delivery.
- A live train-tracking map with real-time ETA, refreshed every 60 seconds.
- An automated delay/disruption notification system meeting the 2-minute SLA.
- A sprint review demo covering the end-to-end booking-to-boarding flow.
- A refined and re-prioritized product backlog ready for Sprint 2 planning.